

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI



CONTACT & PROFILES

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📄 Google Developer Program

AI & LLM STACK

Google Gemma 4 (12B) ■■■■■
Gemini 2.5 / 1.5 Pro ■■■■■
Google TabFM (Tabular) ■■■■■
LangGraph Multi-Agents ■■■■■
Firebase Genkit ■■■■■
Neo4j GraphRAG (N10S) ■■■■■

DATA & GRAPHS

Neo4j / Cypher Graph ■■■■■
PostgreSQL / pgvector ■■■■■
Google BigQuery Data ■■■■ □
Supabase Realtime ■■■■ □
Pandas / NumPy ETL ■■■■■

DEV & MLOPS

Python (3.11+ MLOps) ■■■■■
FastAPI / Next.js 14 ■■■■ □
MLflow & Data Drift ■■■■ □
SHAP Sentinel Audit ■■■■■
Docker & Streamlit ■■■■■
IfcOpenShell (5D BIM) ■■■■ □

SECURITY & AUDIT

◆ Quorum 4-Eyes (KOA+AZIZ)
◆ OKF v0.2 SHA-256 No-LLM
◆ EU AI Act (RSASSA-PSS)
◆ AVSEC (ICAO Annex 17)

LANGUAGES

French (Native / Fluent)
English (Technical / Pro)

KEY ASSETS

◆ Dual Competence AI & Civil Eng.
◆ Aviation Security & Crisis (CCAA)
◆ Mathematical Rigor & Guardrails

◆ EXECUTIVE SUMMARY

Lead AI Engineer & Data Architect (Google Developer Program Member) specializing in sovereign autonomous multi-agents, Neo4j GraphRAG, and deterministic civil engineering software. Founder of Archi Cam AI (official applicant for Google Africa Applied AI Lab), bridging multi-agent orchestration with strict mathematical compliance (BAEL 91, ICAO Annex 17) and production MLOps.

◆ FLAGSHIP AI PROJECTS

Archi Cam AI – *Agentic AI & 5D BIM SaaS*

- ▶ Official Applicant Google Africa Applied AI Lab. 5D BIM & BOQ estimation (local Gemma 4 12B, Gemini, BAEL 91).
- ▶ Automated Excel BOQs in <45s (-99.2% time, MLflow R²=0.9872) and 3D renders via Imagen 3 + ControlNet.

K1-MATHINFO (v3.0.0) – *Sovereign Multi-Agent AI & OKF Certification*

- ▶ Sovereign DMI system (Univ. of Ngaoundéré): 470 theses, 19 M1 projects, Neo4j graph of 1,366 nodes / 3,833 edges.
- ▶ 6-agent LangGraph network, hybrid retrieval (RRF k=60 + Cross-Encoder), SHA-256 No-LLM cert, 77/77 tests (100%).

Sovereign.BI Agentic – *Enterprise Security & Agentic BI*

- ▶ Sovereign natural language query engine for enterprise SQL/Graph databases (PostgreSQL pgvector, Neo4j, <5s latency).
- ▶ Enforces dynamic ABAC security guardrails and explainability certified via SHAP Sentinel game theory auditor.

Dataset Automator & VigieSahel – *MLOps Pipeline & Climate AI Impact*

- ▶ Dataset Automator: TabFM & BigQuery DataFrames MLOps factory with continuous Data Drift tracking (KS/PSI) and EU AI Act.
- ▶ VigieSahel: Predictive ML for epidemic forecasting & crop sowing optimization (-35% loss, XGBoost R²>94%, Supabase).

◆ PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant (2025 – Present)

Independent Projects & Enterprises | Douala, CM

- ▶ Guiding enterprises in deploying sovereign private AI agents, Neo4j GraphRAG pipelines, and PostgreSQL databases.
- ▶ High-dimensional exploratory data analysis, predictive modeling, and executive interactive decision dashboards.

Aviation Security Officer (AVSEC) (2018 – Present)

CCAA (Cameroon Civil Aviation Authority)

- ▶ Operational threat assessment, secure access control, and regulatory compliance audits (ICAO Annex 17).
- ▶ Critical aviation security reporting and field emergency operational crisis response coordination.

◆ EDUCATION & CERTIFICATIONS

Master of Science in Applied Artificial Intelligence (2025 – 2027) [In Progress]

University of Ngaoundéré

Excellence Curriculum (5 Blocs): Machine Learning & Bayesian Statistics, Data Engineering & Neo4j Knowledge Graphs, Computer Vision & Robotics/HCI, Cybersecurity & Blockchain, Production Sovereign MLOps Deployment.

Bachelor of Science in Civil Engineering (Building Option) (2015 – 2016)

ISTDI / IUC Douala

Structural calculations (BAEL 91), quantity surveying, and construction project management.

◆ HONORS & APPLIED AI RECOGNITIONS

- ◆ **Google Africa Applied AI Lab (Accra, 2026):** Official candidacy supported by Archi Cam AI platform.
- ◆ **Google Cloud #AllThingsAgentic Hackathon:** Dataset Automator v4.0 (TabFM, WIT, bigframes, MCT).